



Hunters Hill
High School

Student Number

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2024 TRIAL EXAMINATION

Mathematics Extension 2

General

Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/ or calculations

Total marks:

100

Section I – 10 marks (pages 3–7)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 8–15)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 Marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1. Given $z = 3 - 2i$ and $w = 1 + i$, then $\frac{z}{w}$ in simplest form is
- (A) 1
- (B) $5 + i$
- (C) $\frac{1}{\sqrt{2}} - \frac{5}{\sqrt{2}}i$
- (D) $\frac{1}{2} - \frac{5}{2}i$
2. Consider the statement:
'If n is even, then if n is a multiple of 5, then n is a multiple of 10.'
Which of the following is the negation of this statement?
- (A) n is odd and n is not a multiple of 5 or 10
- (B) n is even and n is a multiple of 5 but not a multiple of 10
- (C) If n is even, then n is not a multiple of 5 and n is not a multiple of 10
- (D) If n is odd, then if n is not a multiple of 5 then n is not a multiple of 10

3. If $\omega \neq 1$ is a cube of unity, what is the value of $(1 + \omega)^2$?
- (A) ω^3
- (B) ω^2
- (C) ω
- (D) 1
4. Which of the following is obtained when $\frac{1-i}{1+\sqrt{3}i}$ is expressed in modulus-argument form?
- (A) $\frac{1}{\sqrt{2}} \operatorname{cis}\left(\frac{17\pi}{12}\right)$
- (B) $\sqrt{2} \operatorname{cis}\left(\frac{17\pi}{12}\right)$
- (C) $\frac{1}{\sqrt{2}} \operatorname{cis}\left(\frac{7\pi}{12}\right)$
- (D) $\sqrt{2} \operatorname{cis}\left(\frac{7\pi}{12}\right)$
5. Which of the following is the unit vector perpendicular to $2\underset{\sim}{i} + \underset{\sim}{j} - \underset{\sim}{k}$?
- (A) $-2\underset{\sim}{i} + 2\underset{\sim}{j} - 2\underset{\sim}{k}$
- (B) $\frac{1}{\sqrt{3}}\left(-\underset{\sim}{i} + \underset{\sim}{j} - \underset{\sim}{k}\right)$
- (C) $2\underset{\sim}{i} - 3\underset{\sim}{j} - \underset{\sim}{k}$
- (D) $\frac{1}{14}\left(2\underset{\sim}{i} - 3\underset{\sim}{j} - \underset{\sim}{k}\right)$

6. Which integral is necessarily equal to $\int_{-a}^a f(x)dx$?

(A) $\int_0^a (f(x) - f(-x))dx$

(B) $\int_0^a (f(x) - f(a - x))dx$

(C) $\int_0^a (f(x - a) + f(-x))dx$

(D) $\int_0^a (f(x - a) + f(a - x))dx$

7. The displacement x of a particle at time t is given by
 $x = 6 \sin 5t + 8 \cos 5t$.

What is the maximum velocity of the particle?

(A) 5

(B) 10

(C) $10\sqrt{5}$

(D) 50

8. The points A, B and C are collinear, where

$$\overrightarrow{OA} = a\mathbf{i} + 2\mathbf{j} - \mathbf{k}, \quad \overrightarrow{OB} = -2\mathbf{i} + b\mathbf{j} + \mathbf{k}, \quad \overrightarrow{OC} = 3\mathbf{i} - \mathbf{j} + 4\mathbf{k}$$

What are the values of a and b ?

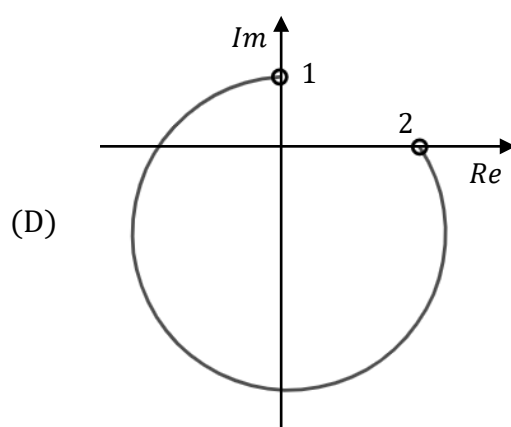
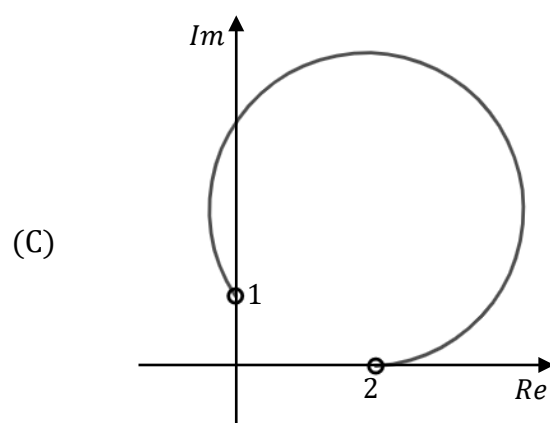
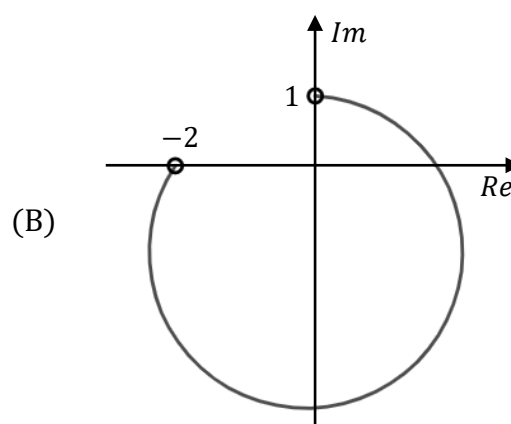
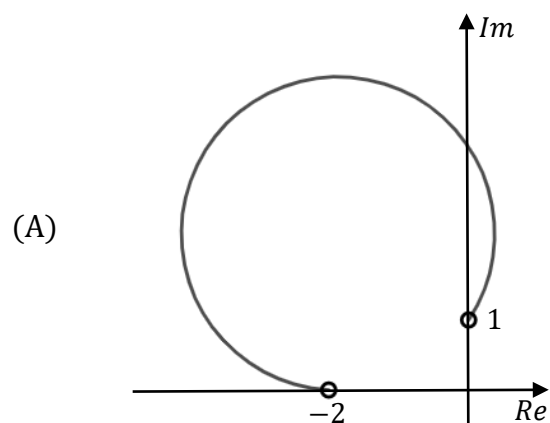
(A) $a = 5, \quad b = -3$

(B) $a = -4, \quad b = \frac{8}{7}$

(C) $a = \frac{-16}{3}, \quad b = \frac{4}{5}$

(D) $a = \frac{1}{3}, \quad b = 8$

9. Which of the following curves represents $\arg\left(\frac{z-i}{z+2}\right) = \frac{\pi}{3}$?



10. The acceleration, $a \text{ ms}^{-2}$, of a particle moving in a straight line is given by $a = 8 - v^2$, where v is the velocity of the particle at any time t . The initial velocity of the particle when at the origin O is 2 ms^{-1} .

The displacement of the particle from O when its velocity is $\sqrt{3} \text{ ms}^{-1}$ is

- (A) $\ln \frac{1}{2}$
- (B) $\frac{1}{2} \ln \left(\frac{4}{5} \right)$
- (C) $\frac{1}{2} \ln \left(\frac{1}{5} \right)$
- (D) $\frac{1}{2} \ln \left(\frac{5}{8} \right)$

End of Section I

Section II

90 Marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Begin each question in a new writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (16 marks) Begin a new Writing Booklet.

(a) Find the square roots of $15 + 8i$. 2

(b) i. Write $1 - \sqrt{3}i$ in mod-arg form. 2

ii. Hence find $(1 - \sqrt{3}i)^5$ in the form $a + ib$. 3

(c) Integrate

i. $\int \sin^3 x \, dx$ 2

ii. $\int \frac{x^3 + 2x + 4}{x + 1} \, dx$ 2

(d) Express $\frac{7x + 11}{(x - 3)(x + 1)^2}$ as a sum of partial fractions over $\mathbb{R}$. 3

(e) The complex numbers $z = 3e^{i\frac{\pi}{3}}$ and $w = 9e^{i\frac{\pi}{4}}$ are given. 2
Find the value of $\frac{z}{w}$, giving the answer in the form $re^{i\theta}$.

End of Question 11

Question 12 (14 marks) Begin a new Writing Booklet.

(a) If n is a positive integer, prove $\sqrt{2 + 6n}$ is always irrational. 3

(b) A polynomial $P(z)$ has the equation $P(z) = z^3 - 6z^2 + kz - 26$, where $k \in \mathbb{R}$. It is known that $2 - 3i$ is a zero of $P(z)$.

i. Find the three complex roots of the polynomial, $P(z)$. 2

ii. Hence, find the value of k . 1

(c) Use an appropriate substitution to evaluate $\int_{\sqrt{7}}^{\sqrt{10}} x^3 \sqrt{x^2 - 6} \, dx$. 3

(d) i. Two people are arguing about the truth of the following statement. 1

$$\exists x \in \mathbb{R} \text{ such that } x^2 = \sqrt{x}$$

Person A claims to prove the above statement by saying that $x = 1$.
Explain why this is sufficient proof.

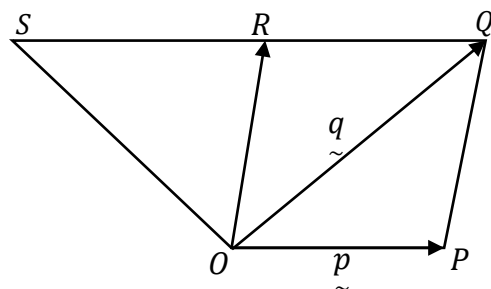
ii. Person B realises his mistake; he'd read the statement as 1

$$\forall x \in \mathbb{R}, \quad x^2 = \sqrt{x}$$

Explain how Person B might have disproved this new statement.

- (e) $OPQS$ is a trapezium in which $\overrightarrow{OP} = \underset{\sim}{p}$ and $\overrightarrow{OQ} = \underset{\sim}{q}$.
 OR is parallel to PQ and $SR : RQ = 2 : 3$.

3



Using vectors, express $\overrightarrow{SP}$ in terms of $\underset{\sim}{p}$ and $\underset{\sim}{q}$.

End of Question 12

Question 13 (15 marks) Begin a new Writing Booklet.

- (a) Find the 5th roots of $32i$. 3
- (b) Integrate $\int \sin^3 \theta \cos^4 \theta d\theta$. 3
- (c) A sequence is defined by 3
 $a_1 = 5, a_2 = 13$ and $a_n = 5a_{n-1} - 6a_{n-2}$ for integers $n \geq 3$.
Use mathematical induction to prove that $a_n = 2^n + 3^n$.
- (d) Using the substitution $t = \tan \frac{x}{2}$, evaluate $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \operatorname{cosec} x dx$. 4
- (e) Find the vector equation of the line passing through points A and B , where 2
 $A = (2, -1, 3)$ and $B = (1, 4, 2)$.

End of Question 13

Question 14 (15 marks) Begin a new Writing Booklet.

- (a) An object moving in a straight line according to the equation

$$x = 6.5 + 5 \sin 3t + 12 \cos 3t.$$

where x is the displacement in metres and t is the time in seconds.

- (i) Prove that the object is moving in simple harmonic motion by showing that x satisfies an equation of the form $\ddot{x} = -n^2(x - c)$. 2
- (ii) When is the displacement of the object zero for the first time? 3

- (b)
- $\tilde{r}_1$
- and
- $\tilde{r}_2$
- are two lines with vector equations:

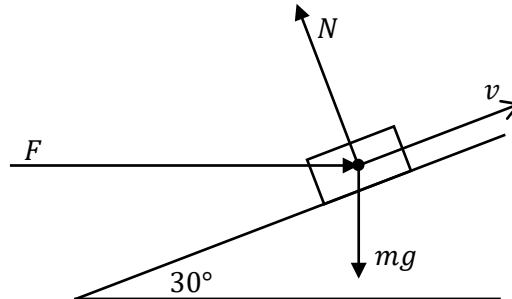
$$\begin{aligned}\tilde{r}_1 &= \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \\ \tilde{r}_2 &= \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \\ &\text{where } \lambda, \mu \in \mathbb{R}.\end{aligned}$$

- (i) Show that these two lines intersect. 2
- (ii) Find the angle between the lines. 1
- (iii) Find the shortest distance from the point $P(1, 2, 2)$ to the line 3

$$\tilde{r}_1 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

- (c) An object of 20 kg mass moves on a smooth inclined plane with velocity, $v(t)$, where t is time in seconds. The inclined plane makes an angle of 30° with the horizontal and the normal reaction force is N , as shown in the diagram.

The acceleration due to gravity is 9.8 ms^{-2} , whilst a lateral constant force F is applied in the direction of the upward slope.



- i. Show that the resultant force on the object, parallel to the slope of the plane, is

2

$$R = \frac{1}{2}(\sqrt{3}F - 196).$$

- ii. If the object is initially at rest, determine the minimum lateral force, in Newtons (kg ms^{-2}), required to push the object 1 m up the slope within 1 minute.

2

End of Question 14

Question 15 (15 marks) Begin a new Writing Booklet.

- (a) A plastic toy is released on the surface of the ocean, at which point it immediately sinks. Let gravity be 10 ms^{-2} and the resistance due to the water is proportional to the square of the velocity.

- (i) Explain why the acceleration of the stone can be given by 1

$$a = 10 - \frac{k}{m}v^2.$$

- (ii) Given $\frac{k}{m} = \frac{1}{10}$, show that $v = 10 \left(\frac{e^{2t} - 1}{e^{2t} + 1} \right)$. 4

- (iii) Using $a = v \frac{dv}{dx}$, show that $x = 5 \ln \left(\frac{100}{100 - v^2} \right)$. 2

- (iv) How far does the toy sink in the first 5 seconds? 2

- (b) Sketch the region of the Argand diagram that satisfies the following conditions 3

$$|z - i| \leq |z - 2 + i|$$

$$-\frac{\pi}{6} \leq \text{Arg}(z) \leq \frac{\pi}{6}$$

- (c) Show that $x\sqrt{x} + 1 \geq x + \sqrt{x}$, for $x \geq 0$. 3

End of Question 15

Question 16 (15 marks) Begin a new Writing Booklet.

- (a) i. Let $I_n = \int_1^e x(\ln x)^n dx$. 3

$$\text{Show that } I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}, \quad n = 1, 2, 3, \dots$$

- ii. Hence evaluate I_2 . 2

- (b) A model plane is launched from atop a tree house, 3 m above the ground with a velocity of $5\hat{i} - 3\hat{j} + 4\hat{k} \text{ ms}^{-1}$ where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are all unit vectors in the east, north and vertically up directions, respectively.

The acceleration of the plane due to the combined effects of gravity and air resistance is $-2\hat{i} + \hat{j} - 2\hat{k}$.

- (i) Show that the angle above the horizontal at which the plane is launched is approximately 0.6 radians. 1
- (ii) Find the displacement vector for the plane. 3
- (iii) Find the maximum height reached by the model plane. 1
- (iv) Find the time taken for the plane to land. 1
- (v) At 3 seconds after launch, a sudden wind kicks up, impacting the flight of the plane, applying *additional* acceleration of $\hat{i} + 3\hat{j} - \hat{k}$. 4

The new vector represents changes to the current system of forces acting upon the model plane.

How far from its intended destination will it now land?

End of Examination

HHHS
2024 Trial Exam
Mathematics Extension 2
Solutions and Marking
Guidelines

1. $z = 3 - 2i$
 $w = 1 + i$

$$\frac{z}{w} = \frac{3-2i}{1+i} \times \frac{1-i}{1-i}$$

$$= \frac{3-2i-3i-2}{1+1}$$

$$= \frac{1-5i}{2} \Rightarrow D$$

2. $\neg(P \Rightarrow (Q \Rightarrow R)) = P \wedge \neg(Q \Rightarrow R)$
 $= P \wedge Q \wedge \neg R$

$\Rightarrow B$

3. $(1+w)^2$
but $1+w+w^2=0$ $w \neq 1$
so $1+w = -w^2$

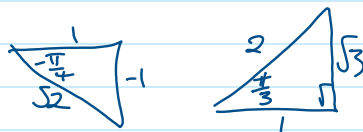
$$(1+w)^2 = (-w^2)^2$$

$$= w^4$$

$$= w^3 \times w$$

$$= w \Rightarrow C$$

4. $\frac{1-i}{4\sqrt{3}}$



$$= \sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right) \div 2 \operatorname{cis} \frac{\pi}{3}$$

$$= \frac{1}{\sqrt{2}} \operatorname{cis}\left(-\frac{\pi}{4} - \frac{\pi}{3}\right)$$

$$= \frac{1}{\sqrt{2}} \operatorname{cis}\left(-\frac{7\pi}{12}\right) \Rightarrow A$$

5. Let $a = 2i + j - 2k$.

A: $2(-2) + 1(2) - 1(2) = -4 + 2 - 2$

$= 0 \therefore$ perp.

but $|a| = \sqrt{(-2)^2 + 2^2 + (-2)^2}$

$= 2\sqrt{3} \neq 1$ so not a unit vector

$$B: -1(2) + 1(1) - 1(-1) = -2 + 1 + 1$$

$$= 0$$

$$|R| = \frac{1}{\sqrt{3}} \sqrt{(-1)^2 + 1^2 + (-1)^2}$$

$$= \frac{\sqrt{3}}{\sqrt{3}}$$

$$= 1$$

$\Rightarrow B$

6. $\int_{-a}^a f(x) dx = F(a) - F(-a)$

Consider $\int_0^a f(x) dx$

let $u = a - x \Rightarrow x = a - u, dx = -du$

$$\begin{aligned} \int_0^a f(x) dx &= \int_0^a f(a-u) du \\ &= \int_a^0 f(a-u) du \\ &= \int_0^a (a-x) dx \end{aligned}$$

Consider $\int_{-a}^0 f(x) dx$

let $u = x + a, \Rightarrow x = u - a, dx = du$

$$\begin{aligned} \int_{-a}^0 f(x) dx &= \int_{-a}^0 f(u-a) du \\ &= \int_0^a f(x-a) dx \end{aligned}$$

$$\therefore \int_{-a}^a f(x) dx = \int_0^a (f(x-a) + f(a-x)) dx$$

$\Rightarrow D$

7. $x = 6 \sin 5t + 8 \cos 5t$

$$\begin{aligned} v &= 30 \cos 5t - 40 \sin 5t \\ &= 50 \cos(5t + \epsilon) \end{aligned}$$

amplitude is max velocity

$\Rightarrow D$

8.

$$\vec{AB} = k \vec{CB}$$

$$\begin{pmatrix} -2-a \\ b-2 \\ 1-(-1) \end{pmatrix} = \begin{pmatrix} 3+2 \\ -1-b \\ 4-1 \end{pmatrix} k \quad \begin{matrix} -i \\ -ii \\ -iii \end{matrix}$$

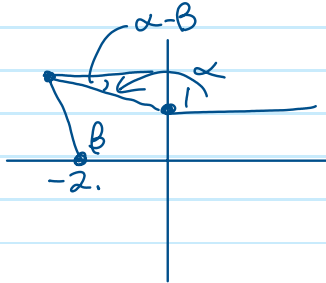
from (iii) $k = \frac{2}{3}$

$$\begin{aligned} \textcircled{1} \quad b-2 &= \frac{2}{3}(-1-b) \\ 3b-6 &= -2-2b \\ 5b &= 4 \\ b &= \frac{4}{5} \end{aligned}$$

$$\begin{aligned} \textcircled{1} \quad -2-a &= \frac{2}{3}(3+2) \\ -a &= \frac{10}{3}+2 \\ a &= -\frac{16}{3} \end{aligned}$$

$\Rightarrow C$

9.



$\Rightarrow A$

10.

$$\begin{aligned} a &= 8-v^2 \\ v \frac{dv}{dx} &= 8-v^2 \end{aligned}$$

$$-\frac{1}{2} \int_2^{\sqrt{3}} \frac{v dv}{8-v^2} = \int_0^x dx$$

$$-\frac{1}{2} \left[\ln|8-v^2| \right]_2^{\sqrt{3}} = x$$

$$x = -\frac{1}{2} \left(\ln|8-3| - \ln|8-4| \right)$$

$$= -\frac{1}{2} \ln\left(\frac{5}{4}\right)$$

$\Rightarrow B$

Question 11

Sunday, 4 August 2024 5:02 PM

a) let $(x+iy)^2 = 15+6i$

$$x^2 - y^2 + 2xyi = 15 + 6i$$

equating Re + Im

$$x^2 - y^2 = 15$$

$$xy = 4$$

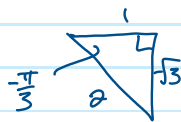
by observation, $x=4, y=1$

$$\therefore \sqrt{15+6i} = \pm(4+i)$$

1- squares $x+iy$

2- answers

b) i $1 - \sqrt{3}i = 2\text{cis}\left(-\frac{\pi}{3}\right)$



1- draws appr. Δ
- correct mod.
2- corrections

ii $(1 - \sqrt{3}i)^5 = \left[2\text{cis}\left(-\frac{\pi}{3}\right)\right]^5$

$$= 2^5 \text{cis}\left(-\frac{5\pi}{3}\right)$$

$$= 32 \text{cis} \frac{\pi}{3}$$

$$= 32 \cos \frac{\pi}{3} + i 32 \sin \frac{\pi}{3}$$

$$= 16 + 16\sqrt{3}i$$

1- applies de Moivre's

2- expands
- writes angle as $\frac{\pi}{3}$

3- correct ans

c) i $\int \sin^3 x dx = \int \sin^2 x \sin x dx$

$$= \int (1 - \cos^2 x) \sin x dx$$

$$= \int (\sin x + \cos^2 x (-\sin x)) dx$$

$$= -\cos x + \frac{\cos^3 x}{3} + C$$

1- splits $\sin^2 \cdot \sin$

2- correct ans
(ignore +C)

ii $\int \frac{x^3 + 2x + 4}{x+1} dx$

$$\begin{array}{r} x^2 - x + 3 \\ x+1 \overline{) x^3 + 2x + 4} \\ \underline{x^3 + x^2} \\ -x^2 + 2x \\ \underline{-x^2 - x} \\ 3x + 4 \\ \underline{3x + 3} \\ 1 \end{array}$$

$$\int \frac{x^3 + 2x + 4}{x+1} dx = \int (x^2 - x + 3 + \frac{1}{x+1}) dx$$

1- performs a step of polynomial div.

2- ans.

$$= \frac{x^3}{3} - \frac{x^2}{2} + 3x + \ln|x+1| + C$$

d). let $\frac{7x+11}{(x-3)(x+1)^2} = \frac{A}{x-3} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$

1 - correct partial fractions

$$\therefore 7x+11 = A(x+1)^2 + B(x-3)(x+1) + C(x-3)$$

for $x=3$
 $7(3)+11 = A(3+1)^2$
 $A=2$

2 - finds a value.

for $x=-1$
 $7(-1)+11 = C(-1-3)$
 $C=-1$

for x^2 term
 $0 = A+B$
 $\therefore B=-2$

$$\therefore \frac{7x+11}{(x-3)(x+1)^2} = \frac{2}{x-3} - \frac{2}{x+1} - \frac{1}{(x+1)^2}$$

3 - correct ans.

e). $z = 3e^{i\frac{\pi}{3}}$ $w = 9e^{i\frac{\pi}{4}}$

$$\frac{z}{w} = \frac{3e^{i\frac{\pi}{3}}}{9e^{i\frac{\pi}{4}}}$$

$$= \frac{1}{3} e^{i(\frac{\pi}{3}-\frac{\pi}{4})}$$

$$= \frac{1}{3} e^{i\frac{\pi}{12}}$$

$$1 - e^{i(\frac{\pi}{3}-\frac{\pi}{4})}$$

2 - ans

Question 12

Sunday, 4 August 2024 5:02 PM

a) Assume $\sqrt{2+6n}$ is rational

$$\therefore \sqrt{2+6n} = \frac{p}{q} \text{ where } p, q \in \mathbb{Z} \text{ and } \text{GCD}(p, q) = 1$$

1 - opening statement

$$2+6n = \frac{p^2}{q^2}$$

$$2q^2(1+3n) = p^2 \quad -i$$

2 - shows p even

$$p^2 \text{ is even} \Rightarrow p \text{ is even}$$

$$\text{so let } p = 2k$$

$$\therefore 2q^2(1+3n) = (2k)^2$$

$$q^2(1+3n) = 2k^2$$

$$2k^2 \text{ is even} \Rightarrow q^2(1+3n) \text{ is even}$$

as $1+3n$ may be even or odd,
 q^2 must be even
 $\Rightarrow q$ is even

and p and q have a common factor of 2.

3 - proof by contradiction

$\therefore$ by contradiction, $\sqrt{2+6n}$ is irrational

b) i $p(z)$ has real coefficients, so complex roots exist in conjugate pairs

$2+3i$ is a zero $\Rightarrow 2-3i$ is a zero
 let third root be α .

1 - justifies conjugate z

$$\begin{aligned} \text{product of roots: } \frac{-(-26)}{1} &= (2+3i)(2-3i)\alpha \\ \alpha &= \frac{26}{2^2 + 3^2} \\ &= 2 \end{aligned}$$

$$\therefore \text{roots are } 2+3i, 2-3i, 2+0i$$

2 - correct roots

ii sum of pairs

$$\begin{aligned} k &= 2(2+3i) + 2(2-3i) + (2+3i)(2-3i) \\ &= 8 + 4 + 9 \\ &= 21 \end{aligned}$$

1 - correct value.

c). $I = \int_7^{\sqrt{10}} x^3 \sqrt{x^2 - 6} \, dx.$

let $u = x^2 - 6$ for $x = \sqrt{7}, u = 1$
 $du = 2x \, dx$ $x = \sqrt{10}, u = 4$

1- chooses and differentiates
appr. subst.

$$I = \frac{1}{2} \int_1^4 (u+6) \sqrt{u} \, du$$

2- substitutes completely

$$= \frac{1}{2} \left[\frac{2u^{\frac{5}{2}}}{\frac{5}{2}} + 6u^{\frac{3}{2}} \cdot \frac{2}{3} \right]_1^4$$

$$= \frac{1}{2} \left[\frac{2}{5} 4^{\frac{5}{2}} + 4 \cdot 4^{\frac{3}{2}} - \left(\frac{2}{5} (1)^{\frac{5}{2}} + 4 \cdot (1)^{\frac{3}{2}} \right) \right]$$

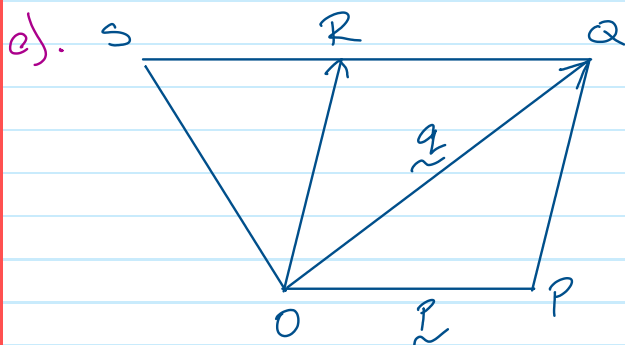
$$= \frac{1}{2} \left(\frac{7}{5} \cdot 32 - \frac{2}{5} - 4 \right)$$

3- correct ans

$$= \frac{101}{5}$$

d) i an example is sufficient to prove a
"there exists" statement

ii a "for all" statement can be disproved
by a counter-example.
e.g. Person B might have
said it's not true for $x=2$



$$\vec{RQ} = \vec{OP} = \underline{p}$$

$$\vec{SR} = \frac{2}{3} \vec{RQ}$$

$$= \frac{2}{3} \underline{p}$$

$$\vec{SQ} = \vec{SR} + \vec{RQ}$$

$$= \frac{2}{3} \underline{p} + \underline{p}$$

$$= \frac{5}{3} \underline{p}$$

1- notes relation
between

- $\vec{SR} + \vec{RQ}$
- $\vec{RQ} \sim \vec{p}$

2- expression for
 $\vec{SQ}$

3- ans

$$\vec{SP} = \vec{SQ} + \vec{QO} + \vec{OP}$$

$$= \frac{5}{3} \underline{p} - \underline{q} + \underline{p}$$

$$= \frac{8}{3} \underline{p} - \underline{q}$$

Question 13

Sunday, 4 August 2024 5:02 PM

a) let $z = 0 + 32i$
 $= 32e^{i\frac{\pi}{2}}$
 $= 32e^{i(\frac{\pi}{2} + 2k\pi)}$

$$z = 2e^{i(\frac{\pi + 4k\pi}{5})}, k = 0, 1, 2, 3, 4, \dots$$

$$= 2e^{i\frac{\pi}{5}}, 2e^{i\frac{5\pi}{5}}, 2e^{i\frac{9\pi}{5}}, 2e^{i\frac{13\pi}{5}}, 2e^{i\frac{17\pi}{5}}$$

$$= 2e^{i(\frac{\pi}{5})}, 2e^{i(\frac{3\pi}{5})}, 2e^{i\frac{\pi}{5}}, 2e^{i\frac{5\pi}{5}}, 2e^{i\frac{9\pi}{5}}$$

1- writes in euler form or equiv.

2- finds exp. for z

3- correct ans

b) $\int \sin^3 \theta \cos^4 \theta d\theta = -\int (1 - \cos^2 \theta) \cos^4 \theta (-\sin \theta) d\theta$
 $= \int (\cos^6 \theta - \cos^4 \theta) (-\sin \theta) d\theta$
 $= \frac{\cos^7 \theta}{7} - \frac{\cos^5 \theta}{5} + C$

1- writes $\sin^2 x$ as $1 - \cos^2 x$

2- appr. intermediate form

3- ans

c) $a_1 = 5, a_2 = 13, a_n = 5a_{n-1} - 6a_{n-2}, n \geq 3$

Prove base cases

$$a_1 = 2^1 + 3^1 = 5$$

$$a_2 = 2^2 + 3^2 = 4 + 9 = 13$$

1- addresses both base cases

Therefore, statement true for $n=1$ and $n=2$

Assume true for $n=3$ to $n=k$

$$a_3 = 2^3 + 3^3$$

$\vdots$

$$a_{k-1} = 2^{k-1} + 3^{k-1}$$

$$a_k = 2^k + 3^k$$

} *

2- stronger assumption

Prove true for $n=k+1$

i.e. prove $a_{k+1} = 2^{k+1} + 3^{k+1}$

$$a_{k+1} = 5a_k - 6a_{k-1}$$

$$= 5(2^k + 3^k) - 6(2^{k-1} + 3^{k-1}) \text{ by } *$$

$$\begin{aligned}
 &= 5(2^k + 3^k) - 3 \cdot 2^k - 2 \cdot 3^k \\
 &= (5-3)2^k + (5-2)3^k \\
 &= 2^{k+1} + 3^{k+1} \quad \text{as required}
 \end{aligned}$$

By the principle of induction, $a_n = 2^n + 3^n$ for all $n \geq 1$.

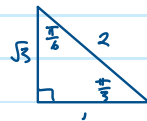
3 - correct proof.

d)

$$t = \tan \frac{x}{2}$$

$$\begin{aligned}
 x = \frac{\pi}{2}, \quad t &= \tan \frac{\pi}{4} = 1 \\
 x = \frac{\pi}{3}, \quad t &= \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}
 \end{aligned}$$

$$\begin{aligned}
 x &= 2 \tan^{-1} t \\
 dx &= \frac{2dt}{1+t^2}
 \end{aligned}$$



$$\int_{\pi/3}^{\pi/2} \operatorname{cosec} x \, dx = \int_{1/\sqrt{3}}^1 \frac{\cancel{1+t^2}}{2t} \cdot \frac{2dt}{\cancel{1+t^2}}$$

$$= \int_{1/\sqrt{3}}^1 \frac{dt}{t}$$

$$= [\ln|t|]_{1/\sqrt{3}}^1$$

$$= \ln 1 - \ln \frac{1}{\sqrt{3}}$$

$$= \frac{1}{2} \ln 3$$

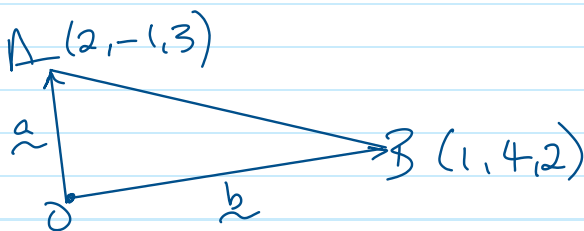
1 - finds dx in terms of t
- change of limits

2 - both the above

3 - correct substitution

4 - ans

e)



$$r = a + \lambda(b-a)$$

$$= \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1-2 \\ 4-(-1) \\ 2-3 \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 5 \\ -1 \end{pmatrix}$$

$$\text{also } r = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 5 \\ -1 \end{pmatrix}$$

1 - find $b-a$ or $a-b$

2 - correct eq.

Question 14

Sunday, 4 August 2024 5:02 PM

a) i $x = 6.5 + 5 \sin 3t + 12 \cos 3t$
 $\dot{x} = 15 \cos 3t - 36 \sin 3t$
 $\ddot{x} = -45 \sin 3t - 108 \cos 3t$
 $= -9 (5 \sin 3t + 12 \cos 3t)$
 $= -9^2 (x - 6.5) \quad \text{as required}$

1 - differentiates and

2 - shown correctly

ii for $x=0$
 $6.5 + 5 \sin 3t + 12 \cos 3t = 0$
 $5 \sin 3t + 12 \cos 3t = -6.5$

let $R \cos(3t - \alpha) = 5 \sin 3t + 12 \cos 3t$
 $R \sin 3t \sin \alpha + R \cos 3t \cos \alpha = 5 \sin 3t + 12 \cos 3t$

$\therefore R \cos \alpha = 12 \Rightarrow R = 13$
 $R \sin \alpha = 5$

1 - attempts to use
auxiliary angle

$\alpha = \tan^{-1}\left(\frac{5}{12}\right)$

$\therefore 13 \cos\left(3t - \tan^{-1}\left(\frac{5}{12}\right)\right) = -6.5$

$\cos\left(3t - \tan^{-1}\left(\frac{5}{12}\right)\right) = -\frac{1}{2}$

2 - correct form
cos/sin

as $\cos \theta < 0$ θ in 2nd + 3rd quadrants
as $\cos \frac{\pi}{3} = \frac{1}{2}$

then $3t - \tan^{-1}\left(\frac{5}{12}\right) = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}, \dots$
 $= \frac{2\pi}{3}, \frac{4\pi}{3}$

$3t = \frac{2\pi}{3} + 0.39479$

$3t = 2.48919$
 $t = 0.82973 \text{ seconds}$

3 - ans

b) i let $\underline{r}_1 = \underline{r}_2$
 $\begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$

$$\begin{aligned} \therefore 2 + 0\lambda &= 1 + \mu &\Rightarrow \mu &= 1 \\ 0 - \lambda &= 1 + \mu(2) &\Rightarrow \lambda &= -3 \end{aligned}$$

1- equates to find λ or μ

$$1 + \lambda = -1 - \mu$$

checking values

$$\begin{aligned} \text{LHS} &= 1 + (-3) & \text{RHS} &= -1 - 1 \\ &= -2 & &= -2 \\ & & &= \text{LHS} \end{aligned}$$

$\therefore$ lines intersect

2- confirms in 3rd equation

ii. compare direction vectors

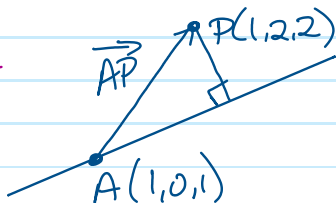
$$\underline{v}_1 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \quad \underline{v}_2 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$$\begin{aligned} \cos \theta &= \frac{\underline{v}_1 \cdot \underline{v}_2}{|\underline{v}_1| |\underline{v}_2|} \\ &= \frac{0(1) - 1(2) + 1(-1)}{\sqrt{0^2 + (-1)^2 + 1^2} \sqrt{1^2 + 2^2 + (-1)^2}} \\ &= \frac{-3}{\sqrt{2} \cdot \sqrt{6}} \\ &= -\frac{\sqrt{3}}{2} \end{aligned}$$

$$\begin{aligned} \theta &= -0.523599 \\ &= 150^\circ / 30^\circ \end{aligned}$$

1- correct angle

iii



$$\vec{AP} = \begin{pmatrix} 1-1 \\ 2-0 \\ 2-1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$$

1- finds $\vec{AP}$
- attempts proj

$$\text{Proj}_{\underline{v}_1} \vec{AP} = \frac{\vec{AP} \cdot \underline{v}_1}{\underline{v}_1 \cdot \underline{v}_1} \underline{v}_1$$

2- correct proj

$$\begin{aligned} &= \frac{0(0) + 2(-1) + 1(1)}{0(0) - 1(-1) + 1(1)} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \\ &= -\frac{1}{2} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \end{aligned}$$

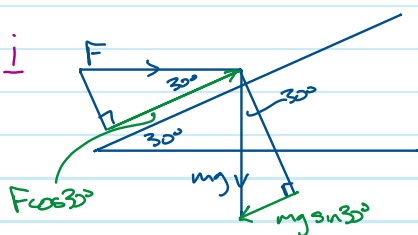
$$\begin{aligned} \text{Shortest distance} &= \left| \vec{AP} - \text{Proj}_{\underline{v}_1} \vec{AP} \right| \\ &= \left| \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} - \left(-\frac{1}{2} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \right) \right| \\ &= \left| \begin{pmatrix} 0 \\ \frac{5}{2} \\ \frac{1}{2} \end{pmatrix} \right| \\ &= \sqrt{0^2 + \left(\frac{5}{2}\right)^2 + \left(\frac{1}{2}\right)^2} \end{aligned}$$

3- ans

$$= \frac{\sqrt{26}}{2} \text{ units}$$

$$= 2.5495093$$

c) i



$$R = F \cos 30^\circ - mg \sin 30^\circ$$

$$= F \left(\frac{\sqrt{3}}{2} \right) - 20(9.8) \frac{1}{2}$$

$$= \frac{1}{2} (\sqrt{3}F - 196)$$

1 - appr. diagram
- finds one parallel component

2 - shown

ii

$$R = ma$$

$$\frac{1}{2} (\sqrt{3}F - 196) = 20a$$

$$a = \frac{1}{40} (\sqrt{3}F - 196)$$

$$\int_0^v dv = \frac{1}{40} (\sqrt{3}F - 196) \int_0^t dt$$

$$v = \frac{1}{40} (\sqrt{3}F - 196) \left[t \right]_0^t$$

$$= \frac{1}{40} (\sqrt{3}F - 196) t$$

$$\int_0^l dx = \frac{1}{40} (\sqrt{3}F - 196) \int_0^{60} t dt$$

$$l = \frac{1}{40} (\sqrt{3}F - 196) \left[\frac{t^2}{2} \right]_0^{60}$$

$$= \frac{3600}{80} (\sqrt{3}F - 196)$$

$$= 45 (\sqrt{3}F - 196)$$

$$F = \frac{\frac{l}{45} + 196}{\sqrt{3}}$$

$$= 113.173493 \text{ N}$$

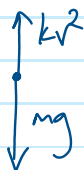
1 - correct exp
for $\checkmark$

2 - ans.

Question 15

Sunday, 4 August 2024 5:02 PM

a) i



Given $F=ma$, the forces due to gravity and resistance are in opposite directions

$$\begin{aligned} ma &= F \\ &= mg - kv^2 \\ a &= g - \frac{k}{m}v^2 \end{aligned}$$

1- sound explanation, noting opposing direction of forces

ii

$$\frac{k}{m} = \frac{1}{10}$$

$$a = 10 - \frac{1}{10}v^2$$

$$\frac{dv}{dt} = \frac{1}{10}(100 - v^2)$$

$$\frac{dv}{100 - v^2} = \frac{1}{10} dt$$

$$\frac{1}{20} \int_0^v \left(\frac{1}{10-v} + \frac{1}{10+v} \right) dv = \int_0^t \frac{1}{10} dt$$

$$\frac{1}{20} \left[-\ln|10-v| + \ln|10+v| \right]_0^v = \frac{1}{10} t$$

$$\ln \left| \frac{10+v}{10-v} \right| - \ln \left| \frac{10}{10} \right| = 2t$$

$$\ln \left| \frac{10+v}{10-v} \right| = 2t$$

$$\frac{10+v}{10-v} = e^{2t}$$

$$\frac{10+v}{10-v} = e^{2t} \Rightarrow v(1+e^{2t}) = 10(e^{2t}-1)$$

$$\therefore v = 10 \frac{e^{2t}-1}{e^{2t}+1}$$

$$\begin{aligned} \frac{A}{10-v} + \frac{B}{10+v} &= \frac{1}{100-v^2} \\ A(10+v) + B(10-v) &= 1 \\ v=10 \Rightarrow A &= \frac{1}{20} \\ v=-10 \Rightarrow B &= \frac{1}{20} \end{aligned}$$

1- initial separation of variable.

2- correct partial fractions.

3- integration

4- result shown

iii.

$$a = 10 - \frac{1}{10}v^2$$

$$\frac{v dv}{dx} = \frac{1}{10}(100 - v^2)$$

$$\frac{1}{2} \int_0^v \frac{2v dv}{100 - v^2} = \frac{1}{10} \int_0^x dx$$

$$\frac{1}{2} \left[\ln|100 - v^2| \right]_0^v = \frac{1}{10} \left[x \right]_0^x$$

1- integration after separation.

$$\frac{1}{2}(\ln 100 - \ln(100 - v^2)) = \frac{1}{10} x$$

$$x = 5 \ln\left(\frac{100}{100 - v^2}\right)$$

iv.

$$t=5 \quad v = 10 \left(\frac{e^{2(5)} - 1}{e^{2(5)} + 1} \right)$$

$$= 9.99092$$

early rounding is problematic

$$x = 5 \ln\left(\frac{100}{100 - 9.99^2}\right)$$

$$= 43.068982 \approx$$

2 - shown

1 - finds v at $t=5$

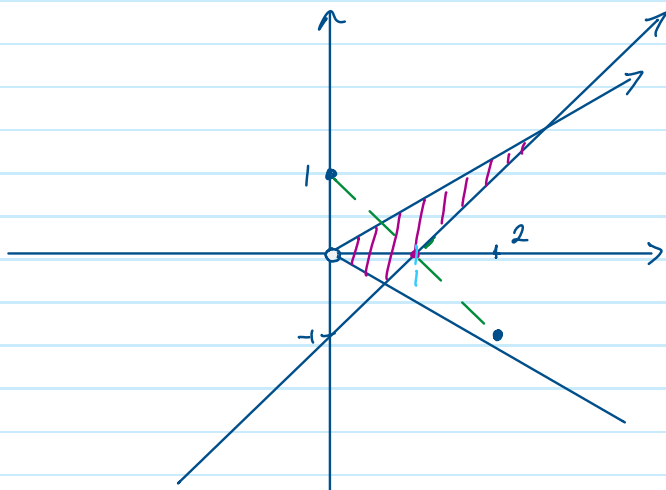
2 - ans

b)

$$|z-1| \leq |z-2+i|$$

$|z-1| = |z-2+i|$ is the perpendicular bisector between $(0,1)$ and $(2,-1)$

$$-\frac{\pi}{6} \leq \arg(z) \leq \frac{\pi}{6} \quad \text{region between two rays}$$



1 - region between r
- perpendicular b

2 - region on li
- region with on
- correct region
open circle

3 - correct con

c)

$$\text{Show } x\sqrt{x+1} > x+\sqrt{x} \quad , \quad x \geq 0$$

$$\begin{aligned} (LHS)^2 &= (x\sqrt{x+1})^2 \\ &= x^3 + 2x\sqrt{x} + 1 \\ &= 2x\sqrt{x} + x^3 + 1 \\ &= 2x\sqrt{x} + (x+1)(x^2-x+1) \end{aligned}$$

1 - squares LHS

$$\begin{aligned} (x-1)^2 &\geq 0 \\ x^2 - 2x + 1 &\geq 0 \\ x^2 + 1 &\geq 2x \end{aligned}$$

2 - uses inequality

$$\begin{aligned} (LHS)^2 &\geq 2x\sqrt{x} + (x+1)(2x-x) \\ &\geq 2x\sqrt{x} + x(x+1) \\ &\geq x^2 + 2x\sqrt{x} + x \\ &\geq (x + \sqrt{x})^2 \\ &\geq (RHS)^2 \end{aligned}$$

3 - proof.

$$\therefore LHS > RHS \quad \text{for } x \geq 0$$

Question 16

Sunday, 4 August 2024 5:02 PM

a) 1

$$I_n = \int_1^e x (\ln x)^n dx$$

$$\text{let } u = (\ln x)^n \quad dv = x dx$$

$$du = n (\ln x)^{n-1} \cdot \frac{1}{x} dx \quad v = \frac{x^2}{2}$$

$$I_n = \left[\frac{x^2}{2} (\ln x)^n \right]_1^e - n \int_1^e \frac{x}{2} (\ln x)^{n-1} dx$$

$$= \frac{e^2}{2} (\ln e)^n - \frac{1^2}{2} (\ln 1)^n - \frac{n}{2} I_{n-1}$$

$$= \frac{e^2}{2} - \frac{n}{2} I_{n-1}$$

1 - chooses u and dv correctly

2 - uses integration by parts

3 - shown

1

$$I_2 = \frac{e^2}{2} - \frac{2}{2} I_1$$

$$= \frac{e^2}{2} - I_1$$

$$I_1 = \frac{e^2}{2} - \frac{1}{2} I_0$$

$$I_0 = \int_1^e x dx$$

$$= \left[\frac{x^2}{2} \right]_1^e$$

$$= \frac{e^2}{2} - \frac{1}{2}$$

1 - finds I_0 or I_2 in terms of I_1

$$I_1 = \frac{e^2}{2} - \frac{1}{2} \left(\frac{e^2}{2} - \frac{1}{2} \right)$$

$$= \frac{e^2}{4} + \frac{1}{4}$$

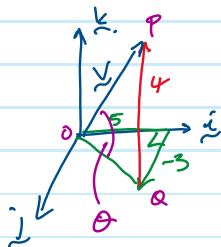
$$I_2 = \frac{e^2}{2} - I_1$$

$$= \frac{e^2}{2} - \left(\frac{e^2}{4} + \frac{1}{4} \right)$$

$$= \frac{e^2}{4} - \frac{1}{4}$$

2 - correct value.

b) i



$$|OQ| = \sqrt{5^2 + (-3)^2}$$

$$= \sqrt{34}$$

$$\theta = \tan^{-1} \frac{4}{\sqrt{34}}$$

$$= 0.6026$$

1 - correctly shown inc. diagram(s)

ii

$$\underline{a} = -2\underline{i} + \underline{j} - 2\underline{k}$$

$$\underline{v} = \int (-2\underline{i} + \underline{j} - 2\underline{k}) dt$$

$$= -2t\underline{i} + t\underline{j} - 2t\underline{k} + \underline{c}$$

$$\text{at } t=0, \underline{v}_0 = 5\underline{i} - 3\underline{j} + 4\underline{k} = \underline{c}$$

$$\text{so } \underline{v} = (5-2t)\underline{i} + (-3+t)\underline{j} + (4-2t)\underline{k}$$

$$\underline{x} = \int ((5-2t)\underline{i} + (-3+t)\underline{j} + (4-2t)\underline{k}) dt$$

$$= (5t - t^2)\underline{i} + (-3t + \frac{t^2}{2})\underline{j} + (4t - t^2)\underline{k} + \underline{c}$$

$$\text{at } t=0, \underline{x}_0 = 3\underline{k} = \underline{c}$$

$$\text{so } \underline{x} = (5t - t^2)\underline{i} + (-3t + \frac{t^2}{2})\underline{j} + (3 + 4t - t^2)\underline{k}$$

1 - integrates

2 - finds $\underline{v}$

3 - finds $\underline{x}$

iii max height when $\underline{v}_k = 0$

$$\therefore 4 - 2t = 0$$

$$t = 2 \text{ s}$$

$$\text{at } t=2, \text{ height} = 3 + 4(2) - (2)^2$$

$$= 7 \text{ m}$$

1 - correct ans

iv plane lands when $\underline{x}_k = 0$

$$\therefore 3 + 4t - t^2 = 0$$

$$t = \frac{-4 \pm \sqrt{4^2 - 4(3)(-1)}}{2(-1)}$$

$$= \frac{-4 \pm \sqrt{28}}{-2}$$

$$= 2 \pm \sqrt{7}$$

$$= 4.6458 \text{ s} \quad \text{taking } +$$

1 - correct ans

v at $t=3$

$$\underline{x}(3) = (5(3) - 3^2)\underline{i} + (-3(3) + \frac{3^2}{2})\underline{j} + (3 + 4(3) - 3^2)\underline{k}$$

$$= 6\underline{i} - \frac{9}{2}\underline{j} + 6\underline{k}$$

$$\underline{v}(4) = (5 - 2(3))\underline{i} + (-3 + 3)\underline{j} + (4 - 2(3))\underline{k}$$

$$= -\underline{i} + 0\underline{j} - 2\underline{k}$$

new acceleration

$$\underline{a} = -2\underline{i} + \underline{j} - 2\underline{k} + (\underline{i} + 3\underline{j} - \underline{k})$$

1 - finds $\underline{x}(3)$

or $\underline{v}(3)$

- original dest.

$$= -\underline{i} + 4\underline{j} - 3\underline{k}$$

$$\underline{r} = -t\underline{i} + 4t\underline{j} - 3t\underline{k} + \underline{c}$$

$$\text{so } \underline{v} = (-1-t)\underline{i} + (4t)\underline{j} + (-2-3t)\underline{k}$$

$$\underline{r} = \left(1 - \frac{t^2}{2}\right)\underline{i} + (2t^2)\underline{j} + \left(1 - 2t - \frac{3t^2}{2}\right)\underline{k} + \underline{c}$$

$$\underline{c} = 6\underline{i} - \frac{9}{2}\underline{j} + 6\underline{k}$$

$$\text{so } \underline{r} = \left(6 - t - \frac{t^2}{2}\right)\underline{i} + \left(2t^2 - \frac{9}{2}\right)\underline{j} + \left(6 - 2t - \frac{3t^2}{2}\right)\underline{k}$$

2 - updated $\underline{v}$

- $\underline{r}(3)$ or $\underline{v}(3)$
+ intend

Intended dest. at ± 4.6458

$$(x, y) = (1.6458, -3.14575)$$

3 - $\underline{v}$ + intended

$$\text{New time of flight: } 6 - 2t + \frac{3t^2}{2} = 0$$

$$12 - 4t - 3t^2 = 0$$

$$t = 1.4415$$

$$\text{New dest. } (x, y) = (5.1296, -3.28557)$$

distance from intended

$$d = \sqrt{(5.1296 - 1.6458)^2 + (-3.28557 + 3.14575)^2}$$

$$= 3.4867 \text{ m}$$

4 - ans